WORKED SOLUTION

STEP II 2011, Question 8

Pure • Involute of a Circle

STEP II 2011, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q8

The key idea

At every moment the string is made of two pieces: an arc of the circle, whose length we can read off from the angle, and a straight piece that leaves the circle along a tangent. Since the total length is fixed, the length of the straight piece is whatever is left over, and a tangent is perpendicular to the radius, so B is easy to locate with vectors. For the final area, notice that the whole region swept out is "area under the outer branch, minus area under the inner branch, minus the semicircle".

Solution

The parametric equations

Let $P = (\cos t, \sin t)$ be the point where the string leaves the circle, so t is the angle from the positive x-axis to OP. At the start $P = A = (-1, 0)$, so $t = \pi$; as the string wraps round over the top of the circle, P moves clockwise and t decreases to 0.

The part of the string in contact with the circle is the arc from angle π to angle t. The radius is 1, so this arc has length $\pi - t$. The string has length π, so the straight piece PB has length

$$\pi - (\pi - t) = t.$$

Because the string is taut, PB is tangent to the circle at P, so it is perpendicular to $OP = (\cos t, \sin t)$. The two unit vectors perpendicular to OP are $\pm(\sin t, -\cos t)$. At the start ($t = \pi$) the string points straight up from A, in the direction $(0, 1) = (\sin \pi, -\cos \pi)$, and by continuity the sign stays the same throughout. Hence

$$\overrightarrow{OB} = \overrightarrow{OP} + t\,(\sin t, -\cos t),$$

that is,

$$x = \cos t + t \sin t, \qquad y = \sin t - t \cos t,$$

as required. (Check: $t = \pi$ gives $(-1, \pi)$, the starting position of B, and $t = 0$ gives $(1, 0)$, where B meets the circle.)

The maximum of x

Differentiating, the terms $\pm\sin t$ and $\pm\cos t$ cancel:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t + \sin t + t\cos t = t\cos t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \cos t - \cos t + t\sin t = t\sin t.$$

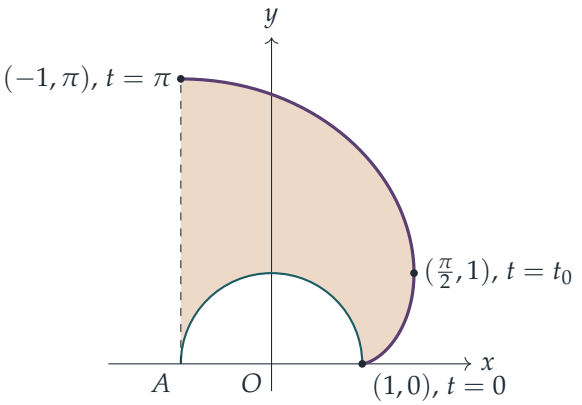
For $0 < t < \pi$, $\frac{\mathrm{d}x}{\mathrm{d}t} = t\cos t$ is positive when $t < \frac{\pi}{2}$ and negative when $t > \frac{\pi}{2}$. So as t runs over $[0, \pi]$, x increases and then decreases, and the maximum is at

$$t_0 = \frac{\pi}{2}, \qquad \text{where } (x, y) = \left(\frac{\pi}{2}, 1\right).$$

The sketch

Two facts help. First, $y > 0$ for $0 < t \leq \pi$: if $t \leq \frac{\pi}{2}$ then $y = \cos t\,(\tan t - t) > 0$ since $\tan t > t$ (for $t = \frac{\pi}{2}$, $y = 1$ directly), and if $t > \frac{\pi}{2}$ then $\cos t < 0$, so both terms of y are non-negative and $\sin t - t\cos t > 0$. Second, $x^2 + y^2 = 1 + t^2$, so B is always outside the circle, by an amount that grows with t. Also $\frac{\mathrm{d}y}{\mathrm{d}t} = t\sin t \geq 0$, so y increases steadily with t.

So the curve starts at $(1, 0)$ (where $\frac{\mathrm{d}y}{\mathrm{d}x} = \tan t = 0$, so it leaves the x-axis horizontally), climbs to its rightmost point $(\frac{\pi}{2}, 1)$, where it is vertical, then sweeps back over the top of the circle to $(-1, \pi)$, where it is horizontal again. This curve is an *involute* of the circle.



The shaded region, between the curve, the initial position of the string (dashed) and the semicircle, is the area swept out by the string.

Area for $\pi \geq t \geq t_0$

On this part of the curve, x runs from $\frac{\pi}{2}$ (at $t = \frac{\pi}{2}$) down to -1 (at $t = \pi$), and $y > 0$. So the area between this arc and the x-axis is

$$\int_{-1}^{\pi/2} y\,\mathrm{d}x = \int_{\pi}^{\pi/2} y\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \int_{\pi/2}^{\pi} (t^2\cos^2 t - t\sin t\cos t)\,\mathrm{d}t,$$

where we swapped the limits to absorb the minus sign. We need two integrals, both by parts after a double-angle formula. Write $\sin t\cos t = \frac{1}{2}\sin 2t$ and $\cos^2 t = \frac{1}{2}(1+\cos 2t)$. By parts,

$$\int t\sin 2t\,\mathrm{d}t = -\tfrac{1}{2}t\cos 2t + \tfrac{1}{4}\sin 2t,$$
$$\int t^2\cos 2t\,\mathrm{d}t = \tfrac{1}{2}t^2\sin 2t - \int t\sin 2t\,\mathrm{d}t = \tfrac{1}{2}t^2\sin 2t + \tfrac{1}{2}t\cos 2t - \tfrac{1}{4}\sin 2t.$$

At $t=\pi$ we have $\sin 2t = 0$, $\cos 2t = 1$; at $t = \frac{\pi}{2}$, $\sin 2t = 0$, $\cos 2t = -1$. Hence

$$\int_{\pi/2}^{\pi} t\sin t\cos t\,\mathrm{d}t = \tfrac{1}{2}\Big[-\tfrac{1}{2}t\cos 2t\Big]_{\pi/2}^{\pi} = \tfrac{1}{2}\Big(-\frac{\pi}{2} - \frac{\pi}{4}\Big) = -\frac{3\pi}{8},$$
$$\int_{\pi/2}^{\pi} t^2\cos^2 t\,\mathrm{d}t = \tfrac{1}{2}\Big[\tfrac{1}{3}t^3\Big]_{\pi/2}^{\pi} + \tfrac{1}{2}\Big[\tfrac{1}{2}t\cos 2t\Big]_{\pi/2}^{\pi} = \frac{7\pi^3}{48} + \tfrac{1}{2}\Big(\frac{\pi}{2} + \frac{\pi}{4}\Big) = \frac{7\pi^3}{48} + \frac{3\pi}{8}.$$

So the required area is

$$\frac{7\pi^3}{48} + \frac{3\pi}{8} - \Big(-\frac{3\pi}{8}\Big) = \frac{7\pi^3}{48} + \frac{3\pi}{4}.$$

The area swept out by the string

Ask yourself The region under the arc for $t \geq t_0$ includes more than the swept region. What must be removed?

The area just found is everything between the outer arc ($t_0 \leq t \leq \pi$) and the x-axis for $-1 \leq x \leq \frac{\pi}{2}$. The swept region is this minus two pieces lying underneath it: the upper semicircle, of area $\frac{\pi}{2}$, and the sliver between the inner arc ($0 \leq t \leq t_0$) and the x-axis for $1 \leq x \leq \frac{\pi}{2}$. The inner arc lies outside the circle (since $x^2 + y^2 = 1 + t^2 > 1$ for $t > 0$) and below the outer arc, so these pieces do not overlap and the subtraction is legitimate.

On the inner arc x increases with t, so the sliver has area

$$\int_0^{\pi/2} y\,\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \int_0^{\pi/2} (t\sin t\cos t - t^2\cos^2 t)\,\mathrm{d}t.$$

Using the same antiderivatives, now with $t = 0$ as the lower limit (where $\sin 2t = 0$, $\cos 2t = 1$),

$$\int_0^{\pi/2} t\sin t\cos t\,\mathrm{d}t = \tfrac{1}{2}\Big[-\tfrac{1}{2}t\cos 2t\Big]_0^{\pi/2} = \frac{\pi}{8},$$
$$\int_0^{\pi/2} t^2\cos^2 t\,\mathrm{d}t = \tfrac{1}{2}\cdot\frac{\pi^3}{24} + \tfrac{1}{2}\Big[\tfrac{1}{2}t\cos 2t\Big]_0^{\pi/2} = \frac{\pi^3}{48} - \frac{\pi}{8},$$

so the sliver has area $\dfrac{\pi}{4} - \dfrac{\pi^3}{48}$ (about 0.14, which is reassuringly small and positive). Therefore the area swept out by the string is

$$\Big(\frac{7\pi^3}{48} + \frac{3\pi}{4}\Big) - \Big(\frac{\pi}{4} - \frac{\pi^3}{48}\Big) - \frac{\pi}{2} = \frac{8\pi^3}{48} = \frac{\pi^3}{6}.$$

Checks

- There is a slicker route to the last answer. When t decreases by δt, the straight piece of string (length t) turns through angle δt about a point that is essentially fixed, sweeping a thin sector of area about $\frac{1}{2}t^2\,\delta t$. Adding these up gives $\int_0^{\pi} \frac{1}{2}t^2\,\mathrm{d}t = \frac{\pi^3}{6}$, in agreement.
- Numerically: Simpson's rule gives 6.87794 for the area under the outer arc, matching $\frac{7\pi^3}{48} + \frac{3\pi}{4}$, and the shoelace formula applied to the boundary of the shaded region gives $5.16771 \approx \frac{\pi^3}{6}$.

Common slips

- Taking the straight piece to have length $\pi - t$ (that is the arc), or choosing the wrong one of the two tangent directions; checking $t = \pi$ catches both.
- Leaving the limits as t_0 to π without noticing that x decreases along that arc, which gives a negative "area".
- Forgetting the small region under the inner arc, which gives $\frac{7\pi^3}{48} + \frac{\pi}{4}$ for the swept area.

Beyond STEP

The curve traced by the end of a string unwinding from a curve is called an involute, and the curve it unwinds from is its evolute: the centre of curvature of the involute at B is exactly the point P where the string leaves the circle. Involutes of circles give the profile of most gear teeth, because two such teeth roll against each other with a constant speed ratio. The "thin sector" argument in the Checks is the polar-area formula $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ in disguise. For more on curvature and curves of this kind, see Siklos, *Advanced Problems in Mathematics*, and for the careful calculus behind it, Abbott, *Understanding Analysis*.

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